

Motion Sensor With Voice Alarm Using Arduino

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Motion sensor alarms are readily available in the market, but these are expensive. Here is a simple and inexpensive Arduino based motion sensor alarm with

in Fig. 4. Search the PCM library from Library Manager (refer Fig. 5) and install it. Similarly, follow the above steps and install the PIR sensor library.

Coding Arduino

Arduino code (Motion_Detection.ino) is written in the Arduino programming language.

Initialise the libraries in the code and also declare the sensor and indicator pins as given below:

```
#include <PCM.h>
int sensorpin = 4; //sensor pin
int led = 13; //indicator pin
```

Initialise the audio file with encoded samples:

```
const unsigned char sample[] PROGMEM =
{ 126, 127, 127, 126, 127, 126, 127, 127,
126, 126, ..... };
```

Define the pinMode using following instructions:

```
pinMode (sensorpin, INPUT);
pinMode (led, OUTPUT);
```

Create a variable that reads the sensor value in digital form:

```
int sensorValue = digitalRead (sensorpin);
```

Now, create a condition that compares with the sensor value. If the condition is true, it performs the task accordingly. Here the task is to play the sample audio file and turn on the LED indicator (LED1). Otherwise, the condition is false, with no audio alert and LED1 off. The following code performs these tasks:

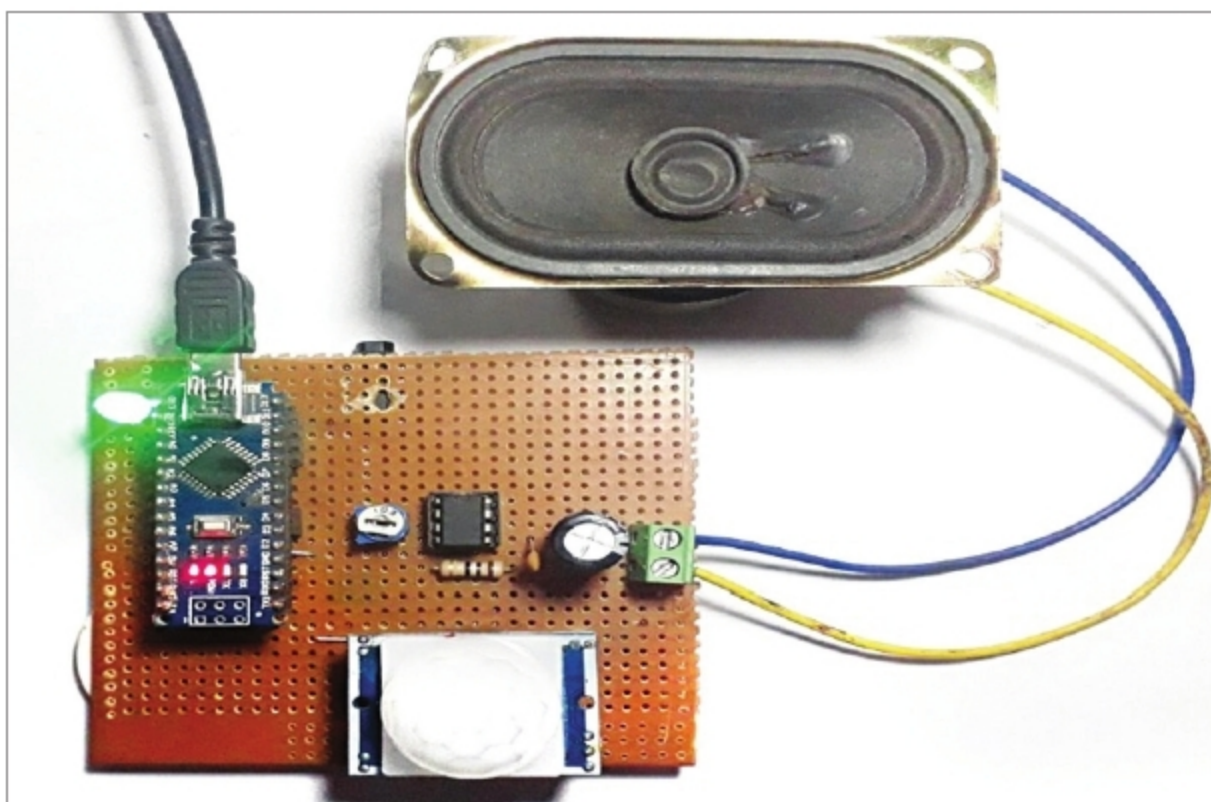


Fig. 1: Author's prototype

voice alert. It can be used for security purposes.

Typically, for outdoor monitoring, motion sensor alarm consists of a PIR (passive infrared) sensor that detects the movement of human beings in the vicinity. The author's prototype is shown in Fig. 1. The block diagram of the motion sensor alarm is shown in Fig. 2.

Circuit and working

The circuit diagram of the motion sensor alarm is shown in Fig. 3. It consists of a PIR motion sensor (MS1), Arduino Nano (Board1), LM386 audio amp (IC1), and 8-ohm speaker. You can use any Arduino board in place of the Arduino Nano board.

Install the required library in Arduino IDE by downloading the library from Sketch→Include library→Manage Libraries options, as shown

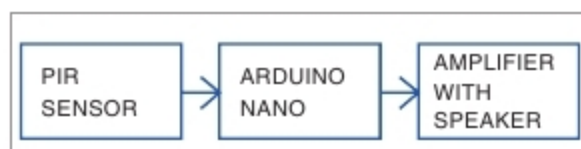


Fig. 2: Block diagram of motion sensor alarm

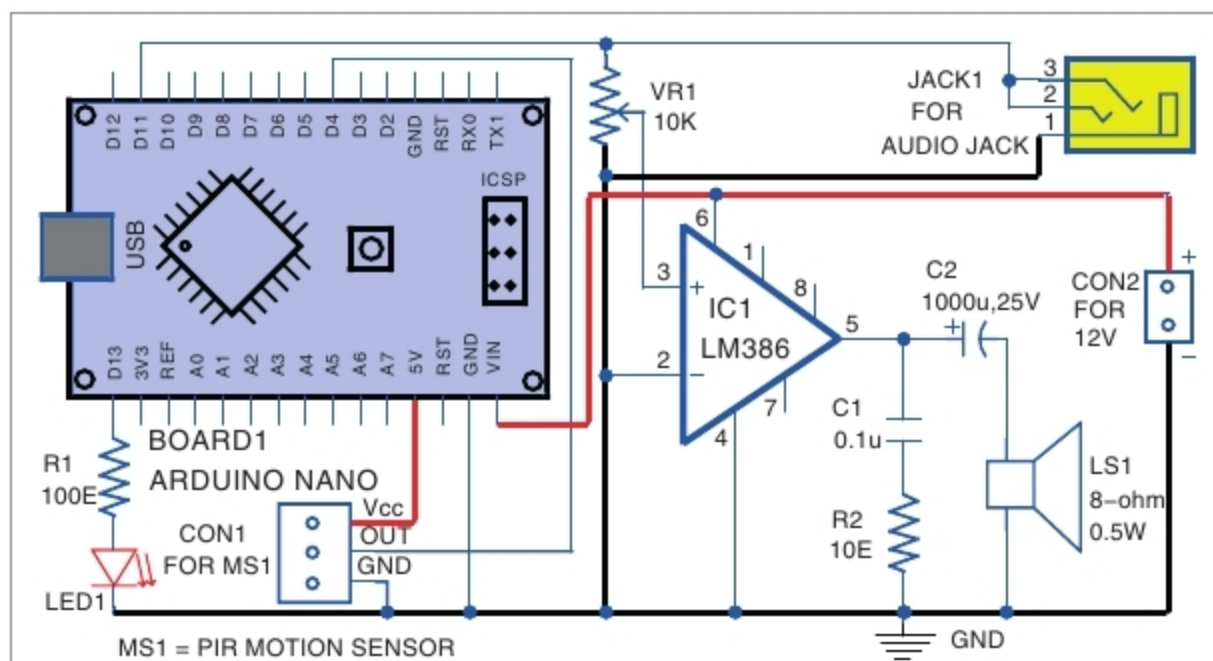


Fig. 3: Circuit diagram of motion sensor alarm